

Solution Requirements

Date	07 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	2 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users securely access the application using authenticated login credentials. (Admin login for dashboard; applicant details validated before processing.)	High
2	Authorization Levels	Admin can manage prediction reports and system settings, while applicants can only submit loan details and view their own prediction results.	High
3	External Interfaces	The application integrates with Machine Learning models, Flask web framework, PDF report generator, Chart.js dashboard, IBM Cloud deployment platform, and local dataset storage.	High
4	Transaction Processing	The system collects applicant details, preprocesses the data, applies feature scaling, predicts loan eligibility using the trained ML model, calculates EMI, and displays the result instantly.	High
5	Reporting	Generates prediction results, applicant summary, approval probability, dashboard statistics, graphical charts, and downloadable PDF reports.	High
6	Business Rules	Loan approval depends on income, credit history, loan amount, employment status, education, dependents, marital status, property area, and other predefined eligibility conditions learned by the ML model.	High
7	Compliance to Laws or Regulations	Applicant data is handled securely, personal information is not shared, and the application follows basic data privacy and ethical AI principles.	Medium
8	Other	Supports EMI calculation, AI-based explanation, responsive interface, dark/light mode, interactive dashboard, and future cloud deployment.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should process applicant information and display prediction results quickly.	Prediction generated in less than 2 seconds
2	Scalability	The application should support increasing users and future integration with banking systems.	Supports 1000+ users with minimal performance degradation
3	Security & Data Privacy	Applicant information should be securely processed and protected from unauthorized access.	Secure data handling with encrypted communication (HTTPS) and role-based access
4	Reliability & Availability	The system should remain available with minimal downtime and provide consistent prediction results.	99.9% availability during deployment
5	Usability & Accessibility	The interface should be intuitive, responsive, mobile-friendly, and easy for users of all experience levels.	Responsive UI compatible with desktop, tablet, and mobile browsers
6	Other	The application should be maintainable, portable, modular, and easy to deploy using Docker and IBM Cloud.	Modular architecture with Flask, Python, and Docker support